

CUSTOMER MILESTONE BRIEF

MS #13 - Sealed Quality Gates

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.13.0
Date	2026-08-14
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #13 - Sealed Quality Gates converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Separates public calibration from private qualification and makes promotion depend on signed, threshold-based evidence.

Why this matters to a customer

Separates public calibration from private qualification and makes promotion depend on signed, threshold-based evidence.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added HMAC-signed, expiring sealed evaluation envelopes with trusted key IDs, exact catalog identity, tamper detection, and fail-closed admission; a plain relabeled catalog is rejected.
- Added opaque holdout case references and privacy-safe receipts that omit private case names, categories, mutation markers, and verifier output from worker artifacts and dashboard APIs.
- Added clean accept-unchanged cases so needless edits are measured independently from mutation repair, including correct no-change and false-acceptance outcomes.
- Added evidence-selection precision, first-attempt repair, private-leak, unauthorized-edit, token, cost, baseline-rejection, and repair metrics under a versioned quality policy.
- Added a promotion-grade quality decision that can qualify only signed sealed evidence meeting every policy threshold; public calibration always remains

non-qualifying.

- Added safety-first evaluation comparison, cross-platform quality-gate launchers, updated schemas, read-only evaluation APIs, and a Quality Control Tower view.
- Added signature, tamper, expiry, wrong-key, clean-case, privacy, policy, comparison, storage, UI, and full-suite regression coverage.

How it differs from earlier work

MS #13 builds on MS #12 (Model Workcell and Context Control). The earlier milestone established its own bounded capability; this milestone adds sealed quality gates while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use sealed quality gates as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.13.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.